

ROAD DESIGN WITH ALLPLAN

PARAMETRIC ROAD DESIGN

ROAD DESIGN WITH ALLPLAN

Free ALLPLAN Ultimate trial
version at allplan.com/ultimate



PARAMETRIC ROAD DESIGN WITH INTUITIVE WORKFLOWS

ALLPLAN offers powerful and easy-to-use modeling functions for road construction projects.

The road model is calculated based on assigned cross section templates and all plans and reports can be automatically derived directly from the model. The parametric concept enables, for example, the gradient of a road axis to be subsequently changed, or the terrain base data to be adjusted.

In addition to high performance, ALLPLAN provides an intuitive user interface, straightforward input workflows and a wide range of functions for model detailing.

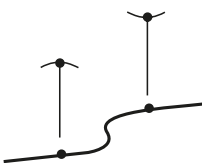
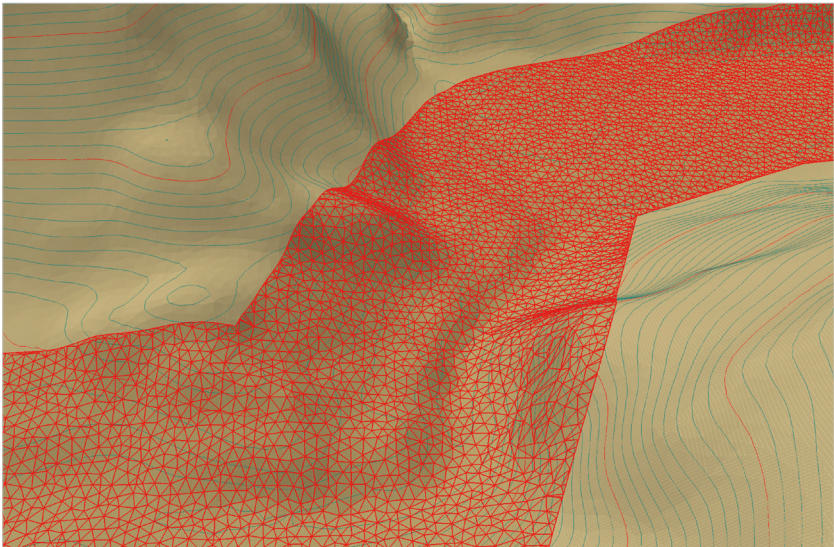
Free webinars at
event.allplanplus.com

EFFICIENT WORKFLOW FOR ROAD PLANNING



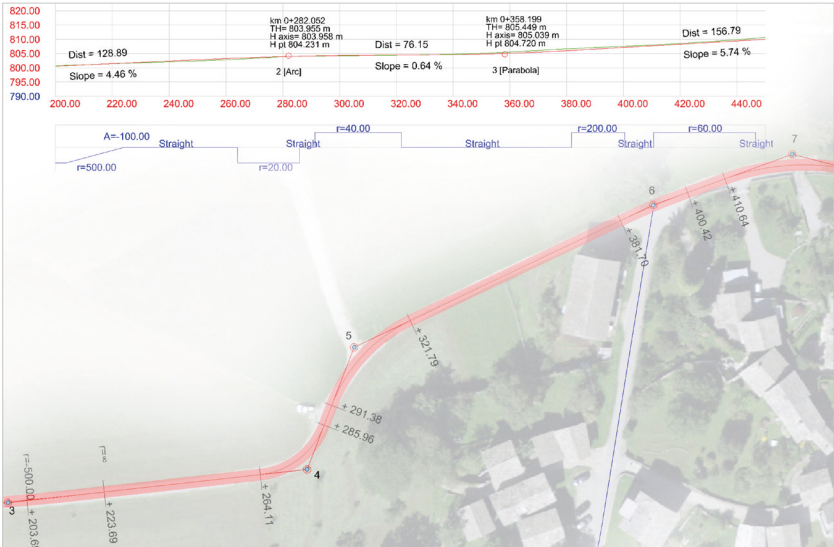
1. POWERFUL TERRAIN MODELING

With the new ALLPLAN terrain module, the most common terrain formats can be loaded, displayed and edited. For the graphical representation of very large terrain models, point clouds can be reduced quantitatively quickly and easily, and terrain edges can be captured and meshed into the terrain surface.



2. PARAMETRIC ROAD AXES

The definition of the horizontal and vertical axis geometry is possible intuitively via the input of tangent intersections or as individual elements. In addition to manual axis input, line elements can be captured as axes and used for further modeling. Due to the parametric concept, the model is immediately recalculated and updated in the event of geometric changes to the axis.

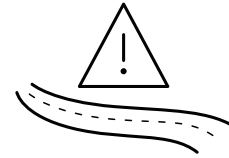
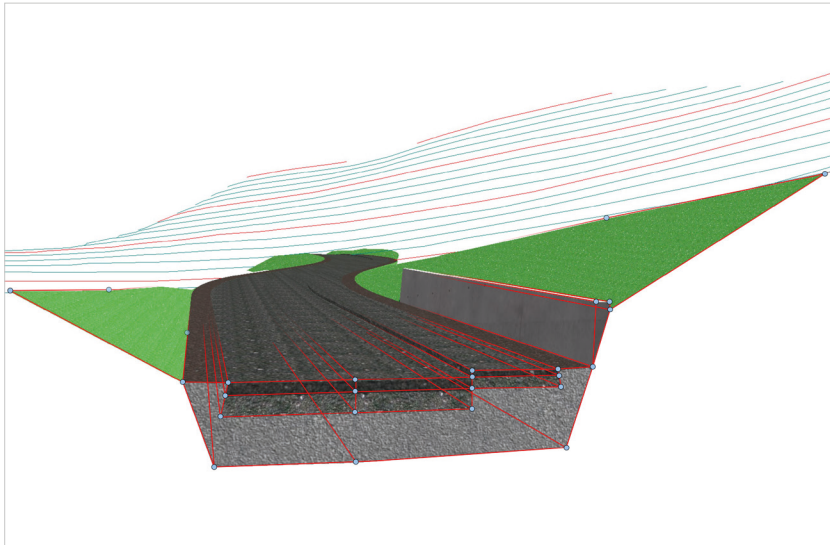


EFFICIENT WORKFLOW FOR ROAD PLANNING



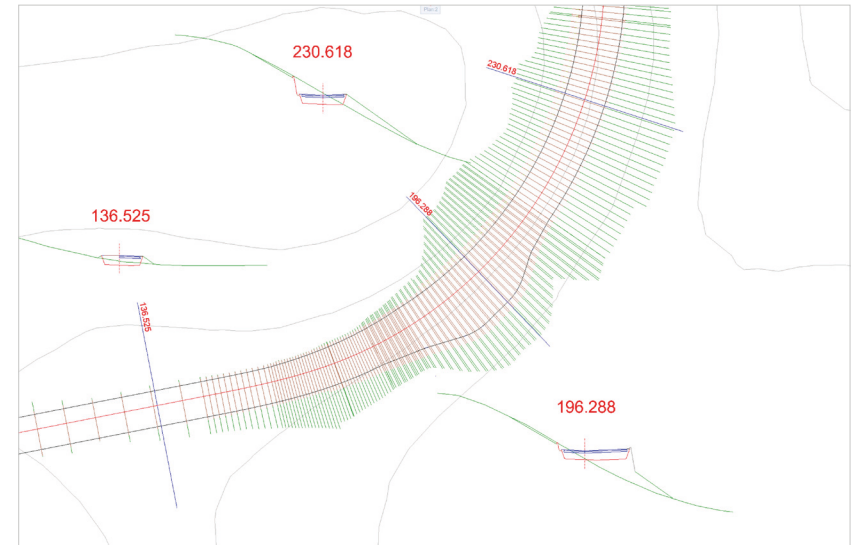
3. TEMPLATE-BASED CROSS-SECTIONAL MODELING

Templates can be used to define the standard profile of the road body. The templates can be defined autonomously for the superstructure, substructure and slope elements and individually combined and assigned to the road axis. The geometric intersection of successive templates is fully automated.

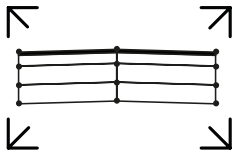


4. AUTOMATED MODEL CHECKING

With the automated model check, the entire road project can be checked easily and quickly. Geometrically faulty profiles or unfulfilled boundary conditions can thus be detected and corrected immediately.

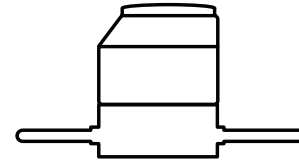
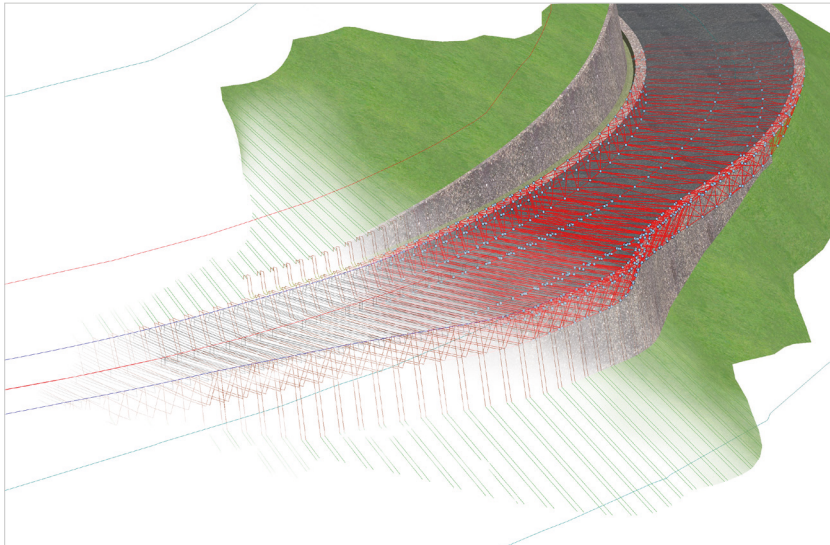


EFFICIENT WORKFLOW FOR ROAD PLANNING



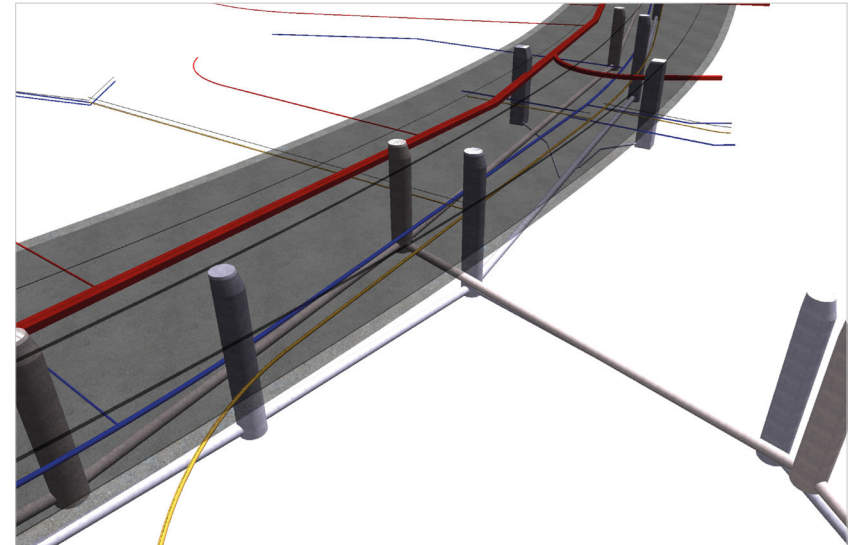
5. FLEXIBLE MODEL DETAILING

The elements of a road cross-section can be easily referenced to corresponding boundary lines or constraint points. It is possible to adapt the characteristic points of the cross-section horizontally and vertically to any geometric situation. The parametric model is recalculated and updated in real time.



6. SEWER AND PIPELINE MANAGEMENT

Utilities within the road, such as manholes and sewer and service lines, can be placed in-situ fully parametrically and adapted according to the course of the road. The line and manhole profiles are selected from a library and can also be easily and subsequently adapted.

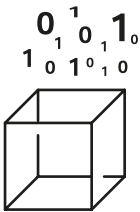
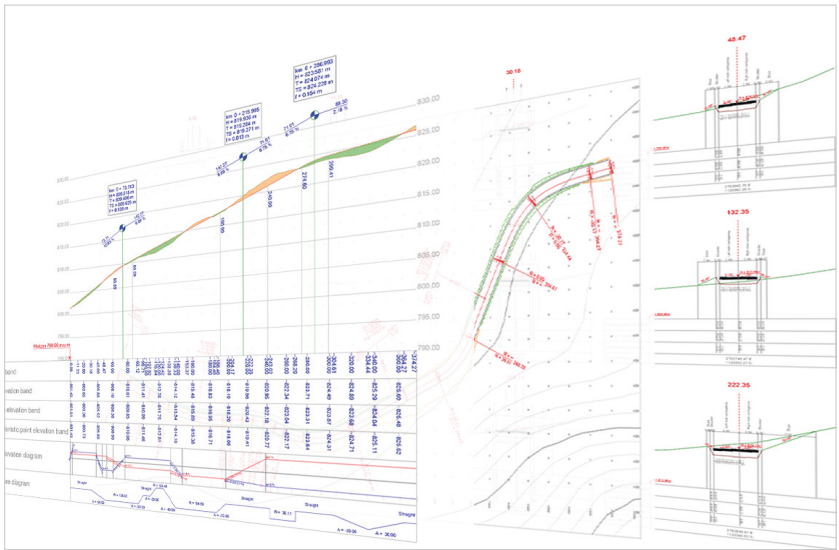


EFFICIENT WORKFLOW FOR ROAD PLANNING



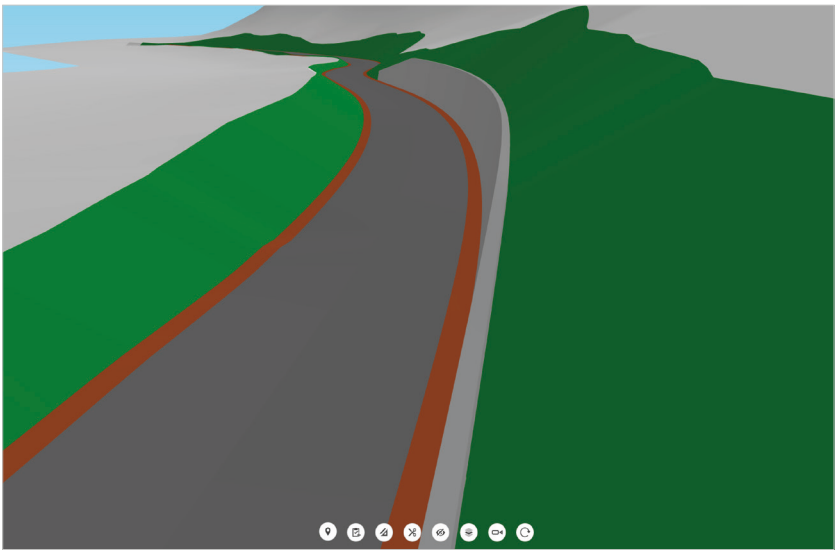
7. PLANS AND REPORTS ON A MODEL BASIS

All plans and reports are created on the basis of the parametric road model. The output-relevant model information is managed in templates, which can be individually defined and adapted at any time. In a few simple steps, site, cross-section and longitudinal profile plans, as well as tabular reports for fill, excavation volumes and staking points can be created.



8. DATA EXCHANGE VIA BIMPLUS

Via BIMPLUS, axis data can be easily exchanged between ALLPLAN and ALLPLAN Civil. In addition to web-based data exchange, project information can be output in LandXML format, the native data format JSON or as a DWG file.



About ALLPLAN

ALLPLAN is a global provider of AEC software with BIM solutions for architecture, structural engineering, detailing, fabrication and construction. True to our "design to build" claim, we provide tools that enable earlier data-driven design decisions, support digital fabrication and leverage information throughout the entire construction process. Integrated cloud technology further optimizes interdisciplinary collaboration on building and infrastructure projects. Our innovative workflows empower architects, engineers, and construction professionals to deliver their projects more productively, safely, and eco-consciously.

Around the world, over 700 dedicated employees continue to write the ALLPLAN success story. Headquartered in Munich, Germany, ALLPLAN is part of the Nemetschek Group – a pioneer for digital transformation in the construction sector.

ALLPLAN is a member of:



Would you like to learn more?

allplan.com/road

ALLPLAN Deutschland GmbH

Konrad-Zuse-Platz 1
81829 München
Deutschland
info@allplan.com
allplan.com

Competence Center

ALLPLAN Infrastructure

info.infra@allplan.com
allplan.com

ALLPLAN
A NEMETSCHKE COMPANY